WORKSHEET · SCRATCH (GR 6–8) INTERMEDIATE: DATA & SIMULATIONS

Lesson 10: A Quiz Engine with Parallel Lists

Module 4 | 45 minutes | Data-driven programs | Grades 6–8 · ages 11–14

Name: _____ Class: _____ Date: _____

Practise what you learned today. Read each task carefully.

1. Key words

Match each word to what it means. Write the letter.

a. parallel lists ___ A. One row of related data.
b. data-driven ___ B. The general code that processes the data.
c. engine ___ C. Behaviour controlled by data, not new code.
d. record ___ D. Lists whose items match up by position.

2. Plan your code

Before you build it: Quiz engine: the questions and answers lists are filled. Set score to 0 and i to 1; repeat (length of questions): ask item i of questions; if answer = item i of answers, change score by 1; change i by 1. Finally say the score. Write or draw the blocks you will use.

3. Out of step

You add a 4th question but forget to add its answer. What happens on question 4? Circle one.

a. A. No answer can match, so it’s always marked wrong
b. B. The quiz crashes
c. C. Scratch adds the answer automatically

4. Data vs code

In a data-driven quiz, the questions are _____ and the loop is the _____. To make a new quiz you change the _____, not the code. (Word bank: data, engine, sprite)

5. Explain your thinking

Give two other apps that are “data-driven” — the same code with different data. Explain what the data is and what the engine does.

How did it go? Colour a face: 😀 I can do it · 🙂 Nearly there · 🤔 I need help

- ☐ I can store a table in parallel lists.
- ☐ I can write one engine loop for any dataset.
- ☐ I can explain data-driven design.